



Research Innovation and Technical Writing CSE414 (Lec.-6)

Research Methodology

Data Preparation, Validation & Experimental
Planning

Objectives

- To understand data preparation steps
- To learn validation techniques
- To design experimental plans
- To define metrics and evaluation strategies

What is Data Preparation?

- Cleaning raw data
- Handling missing values
- Removing duplicates
- Formatting and transformation
- Feature engineering basics



Importance of Data Quality

- Improves model accuracy
- Reduces bias and errors
- Ensures reproducibility
- Supports valid conclusions

Data Validation

- Checking data consistency
- Range and type validation
- Outlier detection
- Cross-checking sources

Common Validation Techniques

- Manual inspection
- Automated scripts
- Statistical checks
- Visualization methods

Experimental Planning Overview

- Define research question
- Select methodology
- Choose dataset and variables
- Plan workflow

Train/Test Split

- Training set: model learning
- Testing set: performance evaluation
- Typical split: 70/30 or 80/20
- Avoid data leakage

Cross Validation

- K-fold validation
- Improves reliability
- Reduces overfitting risk
- Used when data is limited

Evaluation Metrics

- Accuracy, Precision, Recall
- F1-score
- Mean Squared Error
- Confusion matrix

Choosing Metrics

- Depends on problem type
- Classification vs regression
- Business or research goal
- Sensitivity vs specificity trade-off

Evaluation Plan Design

- Define baseline model
- Compare multiple models
- Statistical significance testing
- Reporting standards

Common Mistakes

- Data leakage
- Overfitting to test data
- Poor sampling
- Inconsistent metrics

Summary

- Data quality is critical
- Validation ensures reliability
- Proper planning improves outcomes
- Metrics guide interpretation

Thank You